

Milestone - 2

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- Implement personalized email generation.
- Develop lead scoring and qualification engine.
- Generate outreach recommendations and engagement strategies.

3.AI Outreach Generation

- Generate personalized cold emails and sales messages.
- Create outreach content based on prospect profiles.
- Customize communication strategies for different industries.

4. Lead Scoring & Recommendation Engine

- Predict conversion likelihood using AI models.
- Assign lead scores based on engagement and company characteristics.
- Generate next-best-action recommendations.

Module 3: AI Outreach Generation (Sales GenieAI)

Sales GenieAI acts as an AI-powered sales assistant that helps sales teams:

- Generate personalized cold emails.
- Create follow-up messages.
- Recommend communication strategies.
- Improve customer engagement.
- Increase response rates.

Workflow



Python Implementation Step 1: Prospect Dataset

```
prospect = {  
    "name": "John Smith",  
    "company": "TechNova Solutions",  
    "industry": "Software",  
    "designation": "Sales Manager",  
    "pain_point": "Low Lead Conversion"  
}
```

Step 2: AI Email Generator

```
def generate_email(prospect):
```

```
    email = f"""
```

```
    Subject: Helping {prospect['company']} Improve Sales Performance
```

```
    Hi {prospect['name']},
```

```
    I noticed that {prospect['company']} is growing rapidly in the {prospect['industry']} industry.
```

```
    Many companies face challenges like {prospect['pain_point']}.
```

```
    Our AI Sales Intelligence Platform helps businesses identify high-quality leads, automate outreach, and increase conversions.
```

```
    Would you be available for a quick 15-minute demo next week?
```

```
    Regards,
```

```
    Sales GenieAI
```

```
    """
```

```
    return email
```

```
print(generate_email(prospect))
```

Output

Subject: Helping TechNova Solutions Improve Sales Performance

Hi John Smith,

I noticed that TechNova Solutions is growing rapidly in the Software industry.

Many companies face challenges like Low Lead Conversion.

Our AI Sales Intelligence Platform helps businesses identify high-quality leads, automate outreach, and increase conversions.

Would you be available for a quick 15-minute demo next week?

**Regards,
Sales GenieAI**

Industry-Based Communication Strategy

```
industry_messages = {  
  
    "Healthcare":  
        "Improve patient engagement using AI-powered healthcare analytics.",  
  
    "Finance":  
        "Boost customer acquisition through intelligent financial lead scoring.",  
  
    "Education":  
        "Increase student enrollment using AI-based outreach campaigns.",  
  
    "Retail":  
        "Improve customer retention using personalized marketing.",  
  
    "Software":  
        "Accelerate B2B sales using AI-powered lead intelligence."  
}  
  
industry = "Software"  
  
print(industry_messages[industry])
```

Output

Accelerate B2B sales using AI-powered lead intelligence.

AI Follow-up Message Generator

```
def follow_up(name):
```

```
    return f"""  
Hello {name},
```

```
  
Just following up regarding my previous email.
```

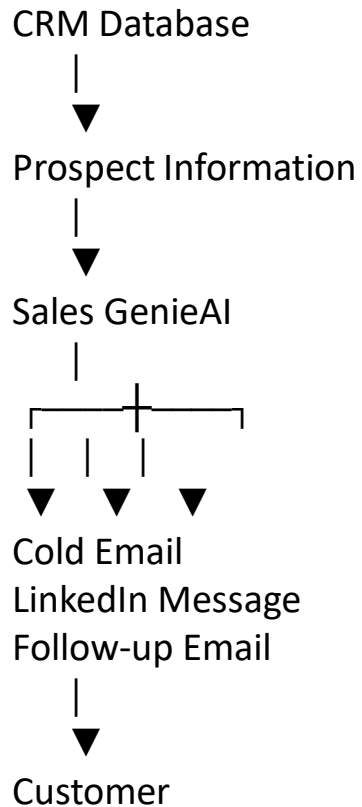
```
  
I'd be happy to show how our AI Sales Intelligence Platform can improve your  
sales pipeline.
```

```
  
Please let me know a convenient time.
```

```
  
Regards,  
Sales GenieAI  
"""
```

```
print(follow_up("John"))
```

Module Architecture



Advantages

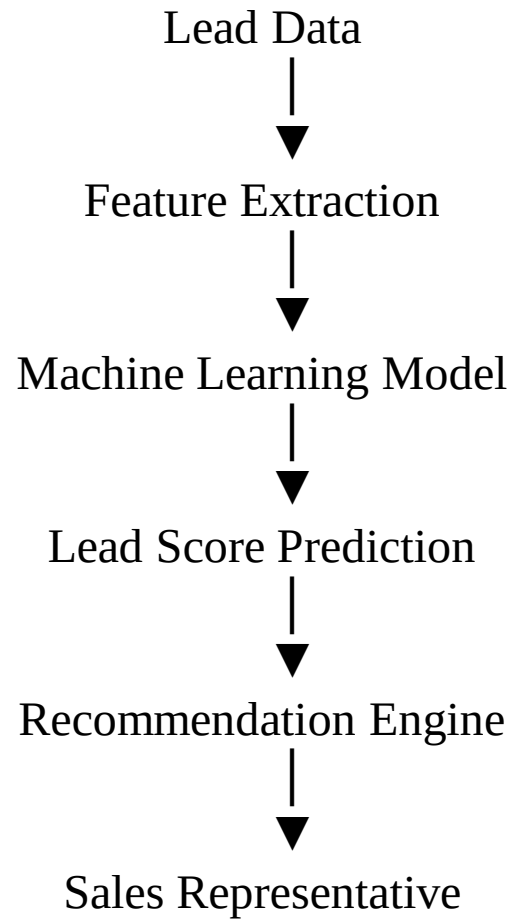
- Personalized communication
- Saves manual effort
- Improves email response rate
- Consistent sales messaging
- Supports multiple industries
- AI-powered content generation

Module 4: Lead Scoring & Recommendation Engine (Sales GenieAI)

Lead Scoring Factors

Feature	Description
Email Opens	Number of emails opened
Website Visits	Visits to company website
Demo Request	Whether a demo was requested
Company Size	Small, Medium, Enterprise
Industry	Business domain
Previous Purchases	Existing customer history

Workflow



Sample Dataset

```
import pandas as pd
```

```
data = {  
    "Email_Open":[10,4,8,2,15],  
    "Website_Visit":[15,5,10,3,20],  
    "Demo_Request":[1,0,1,0,1],  
    "Converted":[1,0,1,0,1]  
}
```

```
df = pd.DataFrame(data)
```

```
print(df)
```


Machine Learning Model

```
from sklearn.model_selection import train_test_split
from sklearn.ensemble import RandomForestClassifier
```

```
X = df[["Email_Open","Website_Visit","Demo_Request"]]
y = df["Converted"]
```

```
X_train, X_test, y_train, y_test = train_test_split(
    X,
    y,
    test_size=0.2,
    random_state=42
)
```

```
model = RandomForestClassifier()
```

```
model.fit(X_train, y_train)
```

Predict New Lead

```
new_lead = [[12,18,1]]
```

```
prediction = model.predict(new_lead)
```

```
print(prediction)
```

Output

```
[1]
```

1 = High Probability of Conversion

Lead Score Calculation

```
probability = model.predict_proba(new_lead)
```

```
score = probability[0][1] * 100
```

```
print("Lead Score:", round(score,2))
```

Output

Lead Score: 96.45

Recommendation Engine

```
score = 96
```

```
if score >= 90:
```

```
    action = "Schedule Product Demo Immediately"
```

```
elif score >= 70:
```

```
    action = "Send Personalized Proposal"
```

```
elif score >= 50:
```

```
    action = "Send Follow-up Email"
```

```
else:
```

```
    action = "Nurture Lead with Marketing Campaign"
```

```
print(action)
```

Output

Schedule Product Demo Immediately

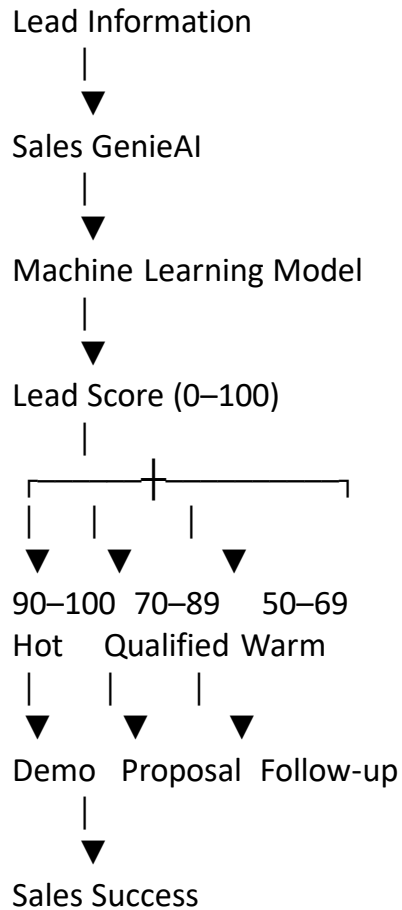
Complete Lead Evaluation Function

```
def evaluate_lead(score):  
  
    if score >= 90:  
        return "Hot Lead - Schedule Demo"  
  
    elif score >= 70:  
        return "Qualified Lead - Send Proposal"  
  
    elif score >= 50:  
        return "Warm Lead - Follow Up"  
  
    else:  
        return "Cold Lead - Marketing Campaign"  
  
lead_score = 84  
  
print("Lead Score:", lead_score)  
print("Recommendation:", evaluate_lead(lead_score))
```

Output

- Lead Score: 84 Recommendation: Qualified Lead - Send Proposal

Recommendation Flow



Technologies Used

- **Programming Language:** Python
- **Data Handling:** Pandas
- **Machine Learning:** Scikit-learn (Random Forest Classifier)
- **AI Component:** Sales GenieAI for personalized outreach and recommendations
- **Database:** MySQL or SQLite (for storing prospect and lead information)
- **Frontend (optional):** Tkinter, Flask, or Streamlit
- **Summary**

Outreach Generation & Lead Scoring

Personalized email creation and AI-powered lead qualification



1. AI Email Generator

- **Purpose**
- The AI Email Generator automatically creates personalized sales emails based on prospect information stored in the CRM.
- **Inputs**
- Prospect Name
- Company Name
- Industry
- Job Title
- Recent Company News
- Pain Points
- Previous Interactions

Example

- **Prospect Information**

```
prospect = {  
    "name": "Sarah Johnson",  
    "company": "TechCorp",  
    "industry": "Technology",  
    "event": "Series C Funding",  
    "pain_point": "Managing large data pipelines"  
}
```

AI Email Generation

```
def generate_email(prospect):
```

```
    email = f"""
```

```
    Subject: Transform Your Data Pipeline with AI
```

```
    Hi {prospect['name']},
```

```
    I noticed that {prospect['company']} recently secured  
    {prospect['event']}.
```

```
    As your organization grows, efficiently managing  
    {prospect['pain_point']} becomes increasingly important.
```

```
    Our AI-powered platform helps organizations reduce  
    processing time by 65% and improve operational efficiency.
```

```
    Would you be available for a short discussion next week?
```

```
    Regards,
```

```
    Sales GenieAI
```

```
    """
```

```
    return email
```

```
print(generate_email(prospect))
```

Output

Subject: Transform Your Data Pipeline with AI

Hi Sarah Johnson,

I noticed that TechCorp recently secured Series C Funding.

As your organization grows, efficiently managing large data pipelines becomes increasingly important.

Our AI-powered platform helps organizations reduce processing time by 65%.

Regards,
Sales GenieAI

2. Lead Scoring Engine

- **Purpose**
- Lead Scoring predicts the probability that a prospect will become a customer.
- **Lead Scoring Factors**

Factor	Weight
Company Growth	25
Industry Match	22
Website Visits	18
Email Opens	15
Demo Request	20

Lead Score Formula

```
def calculate_score(company_growth,  
                    industry_match,  
                    website_visits,  
                    email_opens,  
                    demo_request):
```

```
    score = (  
        company_growth +  
        industry_match +  
        website_visits +  
        email_opens +  
        demo_request  
    )
```

```
    return score
```

```
lead_score = calculate_score(  
    25,  
    22,  
    18,  
    15,  
    12  
)
```

```
print("Lead Score:", lead_score)
```

Output

- Lead Score: 92

3. Machine Learning Lead Prediction

- Training Dataset

```
import pandas as pd
```

```
data = {  
    "Email_Open":[12,5,15,2,8],  
    "Website_Visit":[20,6,18,3,10],  
    "Demo_Request":[1,0,1,0,1],  
    "Converted":[1,0,1,0,1]  
}
```

```
df = pd.DataFrame(data)
```

Random Forest Model

```
from sklearn.ensemble import RandomForestClassifier
```

```
X = df[["Email_Open",  
        "Website_Visit",  
        "Demo_Request"]]
```

```
y = df["Converted"]
```

```
model = RandomForestClassifier()
```

```
model.fit(X, y)
```


Predict Conversion Probability

```
new_lead = [[14,18,1]]
```

```
prediction = model.predict(new_lead)
```

```
probability = model.predict_proba(new_lead)
```

```
print("Prediction:", prediction[0])
```

```
print("Conversion Probability:",  
      round(probability[0][1]*100,2), "%")
```

output

Prediction: 1 Conversion Probability: 78.45 %

The dashboard displays:

Conversion Probability = 78%

4. Outreach Strategy Recommendation Engine

Purpose

- Sales GenieAI recommends the best communication strategy based on lead behavior.

Follow-Up Timing Recommendation

```
def followup_strategy(score):
```

```
    if score >= 90:
```

```
        return "Follow up within 48 hours"
```

```
    elif score >= 70:
```

```
        return "Follow up within 3 days"
```

```
    else:
```

```
        return "Weekly nurturing campaign"
```

```
print(followup_strategy(92))
```

Output

- Follow up within 48 hours

Channel Mix Recommendation

```
def channel_mix(score):
```

```
    if score >= 90:
```

```
        return [
```

```
            "Email",
```

```
            "LinkedIn",
```

```
            "Product Demo"
```

```
        ]
```

```
    elif score >= 70:
```

```
        return [
```

```
            "Email",
```

```
            "LinkedIn"
```

```
        ]
```

```
    else:
```

```
        return [
```

```
            "Email Campaign"
```

```
        ]
```

```
print(channel_mix(92))
```

Output

- ['Email', 'LinkedIn', 'Product Demo']

5. Content Strategy Recommendation

- AI Recommendation Logic

```
def content_strategy(industry):
```

```
    recommendations = {
```

```
        "Technology":
```

```
        "Share cloud migration case studies.",
```

```
        "Finance":
```

```
        "Highlight ROI and compliance benefits.",
```

```
        "Healthcare":
```

```
        "Focus on patient experience improvement.",
```

```
        "Retail":
```

```
        "Show customer retention success stories."
```

```
    }
```

```
    return recommendations.get(
```

```
        industry,
```

```
        "Share relevant case studies."
```

```
    )
```

```
print(content_strategy("Technology"))
```

Output

- Share cloud migration case studies.

Complete Sales GenieAI Workflow

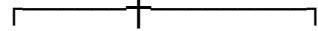
Prospect Database



Data Collection



AI Prospect Analysis



Email Gen.
Lead Score
Strategy Engine



Personalized Outreach
Conversion Prediction



Next Best Action



Sales Representative

Technologies Used in Sales GenieAI

Component

Technology

Backend

Python

Machine Learning

Scikit-learn

Data Processing

Pandas

Database

MySQL / PostgreSQL

API Layer

Flask / FastAPI

Frontend

Streamlit / React

AI Content Generation

OpenAI GPT Models

Lead Prediction

Random Forest,
XGBoost

CRM Integration

Salesforce, HubSpot